MSG No. 191.239 $P6'/m'mm'$ [Type III, hexagonal]

Table 1: Wyckoff site: `1a`, site symmetry: `6'/m'mm'`

No.	position	mapping
1	[0, 0, 0]	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 2: Wyckoff site: `1b`, site symmetry: `6'/m'mm'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16,17,18,19,20,21,22,23,24]

Table 3: Wyckoff site: `2c`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, 0\right]$	[1,2,3,10,11,12,16,17,18,19,20,21]
2	$\left[\frac{2}{3}, \frac{1}{3}, 0\right]$	[4,5,6,7,8,9,13,14,15,22,23,24]

Table 4: Wyckoff site: `2d`, site symmetry: `-6'm2'`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, \frac{1}{2}\right]$	[1,2,3,10,11,12,16,17,18,19,20,21]
2	$\left[\frac{2}{3}, \frac{1}{3}, \frac{1}{2}\right]$	[4,5,6,7,8,9,13,14,15,22,23,24]

Table 5: Wyckoff site: `2e`, site symmetry: `6'mm'`

No.	position	mapping
1	$[0, 0, z]$	[1,2,3,10,11,12,13,14,15,22,23,24]
2	$[0, 0, -z]$	[4,5,6,7,8,9,16,17,18,19,20,21]

Table 6: Wyckoff site: `3f`, site symmetry: `m'mm'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, 0\right]$	[1,4,7,10,14,17,20,23]
2	$\left[0, \frac{1}{2}, 0\right]$	[2,5,8,11,15,18,21,24]
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	[3,6,9,12,13,16,19,22]

Table 7: Wyckoff site: `3g`, site symmetry: `m'mm'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	[1,4,7,10,14,17,20,23]
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	[2,5,8,11,15,18,21,24]

3	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[3,6,9,12,13,16,19,22]`

Table 8: Wyckoff site: `4h`, site symmetry: `3m.`

No.	position	mapping
1	$\left[\frac{1}{3}, \frac{2}{3}, z\right]$	`[1,2,3,10,11,12]`
2	$\left[\frac{2}{3}, \frac{1}{3}, -z\right]$	`[4,5,6,7,8,9]`
3	$\left[\frac{2}{3}, \frac{1}{3}, z\right]$	`[13,14,15,22,23,24]`
4	$\left[\frac{1}{3}, \frac{2}{3}, -z\right]$	`[16,17,18,19,20,21]`

Table 9: Wyckoff site: `6i`, site symmetry: `2'mm'`

No.	position	mapping
1	$\left[\frac{1}{2}, 0, z\right]$	`[1,10,14,23]`
2	$\left[0, \frac{1}{2}, z\right]$	`[2,11,15,24]`
3	$\left[\frac{1}{2}, \frac{1}{2}, z\right]$	`[3,12,13,22]`
4	$\left[\frac{1}{2}, 0, -z\right]$	`[4,7,17,20]`
5	$\left[0, \frac{1}{2}, -z\right]$	`[5,8,18,21]`
6	$\left[\frac{1}{2}, \frac{1}{2}, -z\right]$	`[6,9,16,19]`

Table 10: Wyckoff site: `6j`, site symmetry: `m'2m'`

No.	position	mapping
1	$[x, 0, 0]$	`[1,4,20,23]`
2	$[0, x, 0]$	`[2,5,21,24]`
3	$[-x, -x, 0]$	`[3,6,19,22]`
4	$[-x, 0, 0]$	`[7,10,14,17]`
5	$[0, -x, 0]$	`[8,11,15,18]`
6	$[x, x, 0]$	`[9,12,13,16]`

Table 11: Wyckoff site: `6k`, site symmetry: `m'2m'`

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	`[1,4,20,23]`
2	$\left[0, x, \frac{1}{2}\right]$	`[2,5,21,24]`
3	$\left[-x, -x, \frac{1}{2}\right]$	`[3,6,19,22]`
4	$\left[-x, 0, \frac{1}{2}\right]$	`[7,10,14,17]`
5	$\left[0, -x, \frac{1}{2}\right]$	`[8,11,15,18]`
6	$\left[x, x, \frac{1}{2}\right]$	`[9,12,13,16]`

Table 12: Wyckoff site: 6l, site symmetry: m'm2'

No.	position	mapping
1	$[x,\ 2x,\ 0]$	[1,10,17,20]
2	$[-2x,\ -x,\ 0]$	[2,11,18,21]
3	$[x,\ -x,\ 0]$	[3,12,16,19]
4	$[-x,\ -2x,\ 0]$	[4,7,14,23]
5	$[2x,\ x,\ 0]$	[5,8,15,24]
6	$[-x,\ x,\ 0]$	[6,9,13,22]

Table 13: Wyckoff site: 6m, site symmetry: m'm2'

No.	position	mapping
1	$\left[x,\ 2x,\ \frac{1}{2}\right]$	[1,10,17,20]
2	$\left[-2x,\ -x,\ \frac{1}{2}\right]$	[2,11,18,21]
3	$\left[x,\ -x,\ \frac{1}{2}\right]$	[3,12,16,19]
4	$\left[-x,\ -2x,\ \frac{1}{2}\right]$	[4,7,14,23]
5	$\left[2x,\ x,\ \frac{1}{2}\right]$	[5,8,15,24]
6	$\left[-x,\ x,\ \frac{1}{2}\right]$	[6,9,13,22]

Table 14: Wyckoff site: 12n, site symmetry: ..m'

No.	position	mapping
1	$[x,\ 0,\ z]$	[1,23]
2	$[0,\ x,\ z]$	[2,24]
3	$[-x,\ -x,\ z]$	[3,22]
4	$[x,\ 0,\ -z]$	[4,20]
5	$[0,\ x,\ -z]$	[5,21]
6	$[-x,\ -x,\ -z]$	[6,19]
7	$[-x,\ 0,\ -z]$	[7,17]
8	$[0,\ -x,\ -z]$	[8,18]
9	$[x,\ x,\ -z]$	[9,16]
10	$[-x,\ 0,\ z]$	[10,14]
11	$[0,\ -x,\ z]$	[11,15]
12	$[x,\ x,\ z]$	[12,13]

Table 15: Wyckoff site: 12o, site symmetry: .m.

No.	position	mapping
1	$[x,\ 2x,\ z]$	[1,10]
2	$[-2x,\ -x,\ z]$	[2,11]
3	$[x,\ -x,\ z]$	[3,12]
4	$[-x,\ -2x,\ -z]$	[4,7]
5	$[2x,\ x,\ -z]$	[5,8]
6	$[-x,\ x,\ -z]$	[6,9]
7	$[-x,\ x,\ z]$	[13,22]

continued ...

Table 15

No.	position	mapping
8	$[-x, -2x, z]$	[14,23]
9	$[2x, x, z]$	[15,24]
10	$[x, -x, -z]$	[16,19]
11	$[x, 2x, -z]$	[17,20]
12	$[-2x, -x, -z]$	[18,21]

Table 16: Wyckoff site: 12p, site symmetry: m'..

No.	position	mapping
1	$[x, y, 0]$	[1,20]
2	$[-y, x-y, 0]$	[2,21]
3	$[-x+y, -x, 0]$	[3,19]
4	$[x-y, -y, 0]$	[4,23]
5	$[y, x, 0]$	[5,24]
6	$[-x, -x+y, 0]$	[6,22]
7	$[-x, -y, 0]$	[7,14]
8	$[y, -x+y, 0]$	[8,15]
9	$[x-y, x, 0]$	[9,13]
10	$[-x+y, y, 0]$	[10,17]
11	$[-y, -x, 0]$	[11,18]
12	$[x, x-y, 0]$	[12,16]

Table 17: Wyckoff site: 12q, site symmetry: m'..

No.	position	mapping
1	$\left[x, y, \frac{1}{2}\right]$	[1,20]
2	$\left[-y, x-y, \frac{1}{2}\right]$	[2,21]
3	$\left[-x+y, -x, \frac{1}{2}\right]$	[3,19]
4	$\left[x-y, -y, \frac{1}{2}\right]$	[4,23]
5	$\left[y, x, \frac{1}{2}\right]$	[5,24]
6	$\left[-x, -x+y, \frac{1}{2}\right]$	[6,22]
7	$\left[-x, -y, \frac{1}{2}\right]$	[7,14]
8	$\left[y, -x+y, \frac{1}{2}\right]$	[8,15]
9	$\left[x-y, x, \frac{1}{2}\right]$	[9,13]
10	$\left[-x+y, y, \frac{1}{2}\right]$	[10,17]
11	$\left[-y, -x, \frac{1}{2}\right]$	[11,18]
12	$\left[x, x-y, \frac{1}{2}\right]$	[12,16]

Table 18: Wyckoff site: 24r, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$[-y, x-y, z]$	[2]

continued ...

Table 18

No.	position	mapping
3	$[-x+y,\ -x,\ z]$	[3]
4	$[x-y,\ -y,\ -z]$	[4]
5	$[y,\ x,\ -z]$	[5]
6	$[-x,\ -x+y,\ -z]$	[6]
7	$[-x,\ -y,\ -z]$	[7]
8	$[y,\ -x+y,\ -z]$	[8]
9	$[x-y,\ x,\ -z]$	[9]
10	$[-x+y,\ y,\ z]$	[10]
11	$[-y,\ -x,\ z]$	[11]
12	$[x,\ x-y,\ z]$	[12]
13	$[x-y,\ x,\ z]$	[13]
14	$[-x,\ -y,\ z]$	[14]
15	$[y,\ -x+y,\ z]$	[15]
16	$[x,\ x-y,\ -z]$	[16]
17	$[-x+y,\ y,\ -z]$	[17]
18	$[-y,\ -x,\ -z]$	[18]
19	$[-x+y,\ -x,\ -z]$	[19]
20	$[x,\ y,\ -z]$	[20]
21	$[-y,\ x-y,\ -z]$	[21]
22	$[-x,\ -x+y,\ z]$	[22]
23	$[x-y,\ -y,\ z]$	[23]
24	$[y,\ x,\ z]$	[24]